

TOLUENE**ICSC: 0078 (November 2023)**

Methylbenzene
Toluol
Phenylmethane

CAS #: 108-88-3**UN #: 1294****EC Number: 203-625-9**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Highly flammable. Vapour/air mixtures are explosive. Risk of fire and explosion on contact with strong oxidants.	NO open flames, NO sparks and NO smoking. Closed system, ventilation, explosion-proof electrical equipment and lighting. Prevent build-up of electrostatic charges (e.g., by grounding). Do NOT use compressed air for filling, discharging, or handling. Use non-sparking handtools. NO contact with strong oxidizing agents.	Use foam, powder, carbon dioxide, water spray. In case of fire: keep drums, etc., cool by spraying with water.

AVOID ALL CONTACT!			
	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Sore throat. Cough. Dizziness. Drowsiness. Headache. Nausea. Unconsciousness.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Refer immediately for medical attention.
Skin	Redness. Dry skin.	Protective gloves.	First rinse with plenty of water for at least 15 minutes, then remove contaminated clothes and rinse again. Rinse and then wash skin with water and soap. Refer for medical attention.
Eyes	Redness. Pain.	Wear safety goggles.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.
Ingestion	Aspiration hazard! Burning sensation. Abdominal pain. Vomiting. Further see Inhalation.	Do not eat, drink, or smoke during work.	Rinse mouth. Give nothing to drink. Do NOT induce vomiting. Refer immediately for medical attention. See Notes.

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING

Evacuate danger area! Consult an expert! Personal protection: chemical protection suit and self-contained breathing apparatus. Ventilation. Remove all ignition sources. Do NOT wash away into sewer. Do NOT let this chemical enter the environment. Collect leaking and spilled liquid in sealable containers as far as possible. Absorb remaining liquid in sand or inert absorbent. Then store and dispose of according to local regulations.

STORAGE

Well closed. Fireproof. Separated from strong oxidants. Store in an area without drain or sewer access. Store only in original container.

PACKAGING

According to UN GHS Criteria



DANGER

Highly flammable liquid and vapour

May be fatal if swallowed and enters airways

Causes skin irritation

May cause drowsiness or dizziness

Suspected of damaging fertility or the unborn child

May cause damage to nervous system through prolonged or repeated exposure

Toxic to aquatic life with long lasting effects

Transportation

UN Classification

UN Hazard Class: 3; UN Pack Group: II



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Prepared by an international group of experts on behalf of ILO and WHO, with the financial assistance of the European Commission.

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TOLUENE**ICSC: 0078****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

COLOURLESS LIQUID WITH CHARACTERISTIC ODOUR.

Physical dangers

The vapour mixes well with air, explosive mixtures are easily formed. As a result of flow, agitation, etc., electrostatic charges can be generated.

Chemical dangers

Reacts violently with strong oxidants such as nitric acid and sulfuric acid. This generates fire and explosion hazard. This produces irritating and toxic gases.

Formula: $C_6H_5CH_3$ / C_7H_8

Molecular mass: 92.1

Boiling point: 111°C

Melting point: -95°C

Relative density (water = 1): 0.87

Solubility in water: none

Vapour pressure, kPa at 25°C: 3.8

Relative vapour density (air = 1): 3.1

Relative density of the vapour/air-mixture at 20°C (air = 1): 1.01

Flash point: 4°C c.c.

Auto-ignition temperature: 480°C

Explosive limits, vol% in air: 1.1-7.1

Octanol/water partition coefficient as log Pow: 2.69

Viscosity: 0.68 mm²/s at 20°C**EXPOSURE & HEALTH EFFECTS****Routes of exposure**

The substance can be absorbed into the body by inhalation, through the skin and by ingestion.

Effects of short-term exposure

The substance is irritating to the skin. The substance is mildly irritating to the eyes. The substance may cause effects on the central nervous system. If this liquid is swallowed, aspiration into the lungs may result in chemical pneumonitis. Exposure at high levels could cause cardiac dysrhythmia and unconsciousness.

Inhalation risk

A harmful contamination of the air can be reached rather quickly on evaporation of this substance at 20°C.

Effects of long-term or repeated exposure

The substance defats the skin, which may cause dryness or cracking. The substance may have effects on the central nervous system. Exposure to the substance may increase noise-induced hearing loss. Animal tests show that this substance possibly causes toxicity to human reproduction or development. May cause colour vision impairment.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 20 ppm as TWA; (OTO); A4 (not classifiable as a human carcinogen); BEI issued.

EU-OEL: 192 mg/m³, 50 ppm as TWA; 384 mg/m³, 100 ppm as STEL; (skin)**ENVIRONMENT**

The substance is toxic to aquatic organisms. The substance may cause long-term effects in the aquatic environment. It is strongly advised not to let the chemical enter into the environment.

NOTES

Depending on the degree of exposure, periodic medical examination is suggested.

Use of alcoholic beverages enhances the harmful effect.

Ingestion of this substance, regardless of the amount ingested, can cause aspiration and thus risk of chemical pneumonitis.

The symptoms of chemical pneumonitis do not become manifest until a few hours or even a few days have passed.

ADDITIONAL INFORMATION**EC Classification**

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